Data Analysis Of Global AI Jobs Recruitment

Project Overview and Objective

The **Global AI Jobs 2026** project aims to analyze global employment opportunities in Artificial Intelligence (AI), Machine Learning (ML), and Data Science across different industries and countries. The dataset includes information about job titles, AI specializations, industries, locations, salary categories, educational qualifications, years of experience, required skills, certifications, and job openings. The main objective is to clean and preprocess the data, perform exploratory data analysis (EDA), and create visualizations using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. This analysis helps identify popular AI specializations, in-demand skills, salary patterns, and job opportunities across different locations. The project also aims to generate meaningful business insights and recommendations to help job seekers plan their careers, recruiters improve hiring strategies, and educational institutions align their training programs with industry requirements.

Data Sources

Source Description and Timeline:
OpenDataPlatforms/Kaggle(https://www.kaggle.com/datasets/datascikhan/global-ai-jobs-2026?select=global_ai_jobs_2026.csv)

Year / Timeline: Data collected during 2026

Domain: Artificial Intelligence & Technology Jobs

Problem Statement

- The global AI job market is growing rapidly, making it important to identify in-demand AI specializations, job roles, and employment opportunities across industries and countries.
- Job seekers often face difficulties understanding the technical skills, educational qualifications, certifications, and work experience required for AI-related jobs.
- Analyzing salary categories, job openings, and employment trends across different industries and locations is essential for understanding the global AI job market.
- This project uses Python, data cleaning, exploratory data analysis (EDA), and data visualization to identify job market trends and generate meaningful insights that support career planning, recruitment strategies, and skill development.

Attribute (Column /Features) Details:

S.No.	Attribute / Column Name	Data Type	Description
1	Job_ID	Object (String)	Unique identification number for each job record.
2	Job_Title	Object (String)	Name of the job role.
3	AI_Specialization	Object (String)	Specific AI field associated with the job.
4	Industry	Object (String)	Industry offering the job opportunity.
5	Company	Object (String)	Name of the company offering the job.
6	Country	Object (String)	Country where the job is located.
7	Continent	Object (String)	Continent where the job is available.
8	Education_Required	Object (String)	Educational qualification required for the job.
9	Years_of_Experience	Integer (int64)	Number of years of work experience required.
10	Salary_Category	Object (String)	Salary range or category associated with the job.
11	Skills_Required	Object (String)	Technical and professional skills required.
12	Required_Certifications	Object (String)	Required or preferred certifications.
13	Job_Openings	Integer (int64)	Number of vacancies available.
14	Job_Type	Object (String)	Type of employment, such as Full-time or Part-time.

Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** visualization, and interactive dashboard creation.

Data Pre-Processing (Power Query)

- Load the Dataset: Import the Global AI Jobs 2026 CSV file into Power BI using *Home → Get Data → Text/CSV*.
- Open Power Query Editor: Click Transform Data to open Power Query Editor.

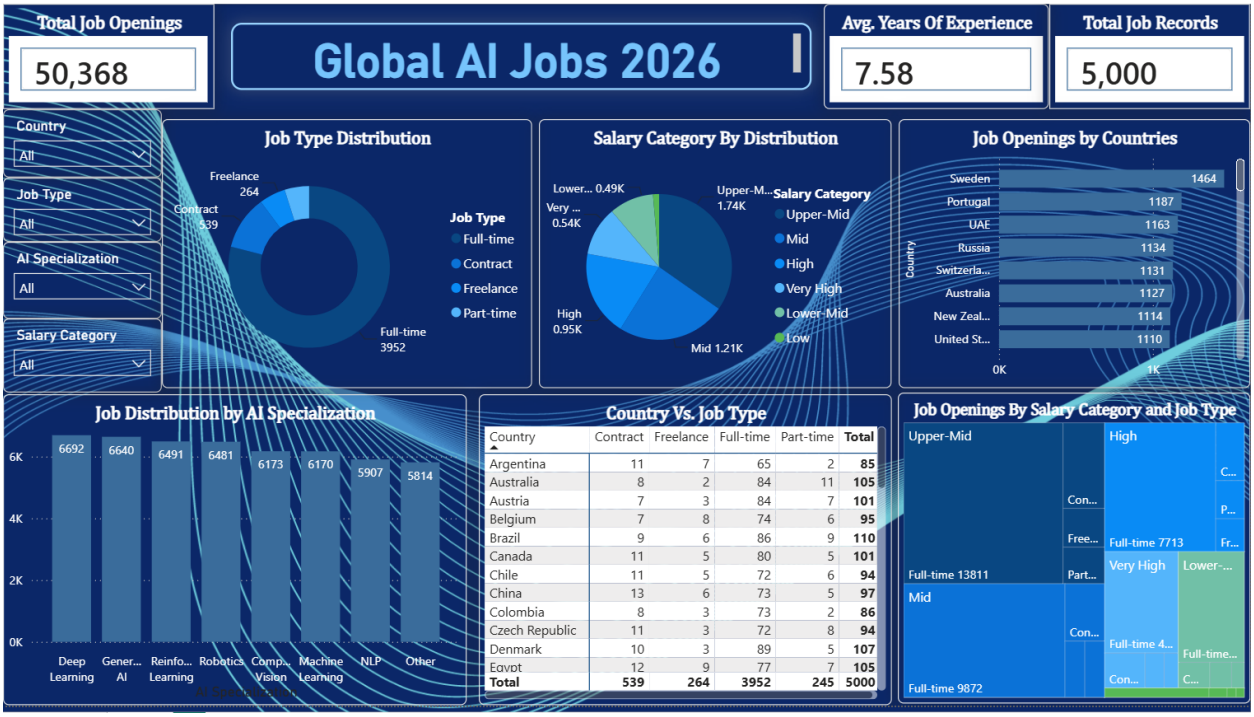
- Handle Missing Values: Identify null or blank values in columns such as Required_Certifications. Replace missing values with an appropriate label or remove records when justified.
- Remove Duplicates: Select the relevant columns or all columns, then choose Home → Remove Rows → Remove Duplicates to eliminate duplicate records.
- Rename Columns: Ensure column names are clear and consistent, such as Years_of_Experience and Job_Openings.

Analysis and Visualizations (Power BI)

Dashboard Features:

- **Job Type Distribution (Donut Chart):** Shows the breakdown of employment terms. **Full-time** dominates with 3,952 positions, followed by Contract (539), Freelance (264), and Part-time (245).
- **Salary Category By Distribution (Pie Chart):** Breaks down compensation tiers, highlighting **Upper-Mid** (1.74K positions) and **Mid** (1.21K positions) as the most frequent categories.
- **Job Openings by Countries (Horizontal Bar Chart):** Ranks the top hiring locations, led by **Sweden** (1,464 openings), Portugal (1,187), and the UAE (1,163).
- **Job Distribution by AI Specialization (Vertical Bar Chart):** Compares demand across engineering disciplines. **Deep Learning** leads with 6,692 listings, closely followed by Generative AI (6,640) and Reinforcement Learning (6,491).
- **Country Vs. Job Type (Data Matrix Table):** A cross-tabular view showing an alphabetical list of countries matched against contract structures, summing up to the 5,000 total record dataset.
- **Job Openings by Salary Category and Job Type (Treemap Chart):** A hierarchical breakdown visualizing combined groupings (e.g., highlighting that Full-time roles within the "Mid" salary bracket account for 13,811 openings).

SCREENSHOT OF DASHBOARD



Country	Contract	Freelance	Full-time	Part-time	Total
Argentina	11	7	65	2	85
Australia	8	2	84	11	105
Austria	7	3	84	7	101
Belgium	7	8	74	6	95
Brazil	9	6	86	9	110
Canada	11	5	80	5	101
Chile	11	5	72	6	94
China	13	6	73	5	97
Colombia	8	3	73	2	86
Czech Republic	11	3	72	8	94
Denmark	10	3	89	5	107
Egypt	12	9	77	7	105
Total	**539**	**264**	**3952**	**245**	**5000**

Insights & Conclusions

Provide the analysis insights:

- **Descriptive Analysis**
  Descriptive analysis summarizes the dataset's core variables using charts and statistical metrics to outline the baseline characteristics of global AI job opportunities.
- **Diagnostic Analysis**
  Diagnostic analysis cross-examines variables to uncover structural patterns, investigating how specific AI specializations, industries, and geographic locations directly influence salary tiers and historical hiring demand.
- **Predictive Analysis**
  Predictive analysis leverages classification models to estimate target metrics like salary categories based on job attributes, though time-based demand forecasting is restricted by the lack of historical timeline data.
- **Prescriptive Analysis**
  Prescriptive analysis translates data-driven findings into actionable workforce recommendations for skill development, strategic recruitment targeting, and career planning tailored to observed market conditions.

Conclusions

The Global AI Jobs 2026 Recruitment in job types, AI specializations, salaries, and country-wise job openings. The analysis identifies full-time employment as the dominant job type and Deep Learning and Generative AI as prominent specializations. These insights help businesses make informed decisions about recruitment, workforce planning, and skill development. Integrating Python with Power BI enables efficient data cleaning, automation, and interactive visualization. Future enhancements using larger datasets and predictive modeling can improve forecasting and support better business decisions.